

Aditya Pande

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AI/ML Engineer specialising in production-grade intelligent systems spanning LLM application development, agentic AI pipelines, real-time telemetry platforms, and multimodal geospatial deep learning. Experienced in end-to-end model development, MLOps, and embedded inference; comfortable bridging research and deployment at scale.

Experience

AI/ML Engineer (Entrepreneur-in-Residence)

Aug 2024 – Present

NeoGeoInfo Technologies Pvt. Ltd.

Hyderabad, India

- Architected multimodal geospatial fusion pipelines integrating satellite imagery, LiDAR, GNSS, aerial, and GPR datasets; achieved 20% improvement in spatial alignment accuracy through CNN-based cross-modal feature learning
- Designed and shipped end-to-end data pipelines covering preprocessing, orthorectification, co-registration, and geometric correction — enabling consistent cross-dataset spatial reasoning for urban planning and infrastructure intelligence
- Built AI-driven spatial analytics workflows (urban planning, utilities mapping, large-scale infrastructure) and evaluated models with RMSE and SSIM metrics to track production performance
- Sole backend and systems engineer on OptiFleet — a real-time IoT fleet-management platform with low-latency GPS/sensor telemetry ingestion, route-optimisation analytics, and fuel-efficiency dashboards (Python, IoT, PostgreSQL, ThingsBoard)

Chief Engineer – Data Acquisition and Telemetry

2021 – 2024

Wrench Welders Racing (Formula Student Team)

India

- Owned the full sensor lifecycle selection, installation, calibration, and real-time DAQ on a Formula Student race car; built live telemetry dashboards (Power BI) that drove data-informed vehicle-setup decisions
- Applied statistical analysis and anomaly detection on race datasets to surface performance trends and reduce lap-time variance across events

Selected Projects

IoT-Enabled Fleet Management & Telemetry Platform (OptiFleet) [product]

2024 – Present

Python, GPS/GNSS, LTE IoT, PostgreSQL, ThingsBoard, Power BI, Computer Vision

- Sole backend/systems engineer for a production fleet-management platform; built scalable, low-latency pipelines ingesting high-frequency GPS, speed, idle-time, and multi-sensor signals from distributed vehicle hardware
- Implemented efficiency analytics for route optimisation, fuel-usage modelling, and idle-time reduction, enabling data-driven operational decisions across the fleet
- Integrated computer-vision-based driver behaviour and safety modules into the telemetry stream, surfacing real-time alerts and event logs

AI-Driven Multi-Source Geospatial Data Fusion

2025

Python, PyTorch, CNNs, GNSS/LiDAR/Satellite/GPR, RMSE/SSIM Evaluation

- Built an end-to-end deep-learning pipeline to fuse heterogeneous geospatial modalities (GNSS, satellite, LiDAR, GPR); used CNN encoders for cross-modal spatial feature learning and alignment
- Quantified model quality with RMSE and SSIM metrics across fusion experiments; delivered actionable spatial-accuracy gains over baseline single-source approaches

3D Printing Defect Detection using Computer Vision

2023 – 2024

Python, YOLOv5, OpenCV, TensorFlow Lite, Raspberry Pi, Edge Inference

- Designed a layer-level defect detection system for FFF/FDM 3D printing; fine-tuned YOLOv5 with data augmentation and hard-negative mining, achieving over 90% detection accuracy and mean IoU exceeding 0.5 across defect categories
- Optimised inference for edge deployment on Raspberry Pi (TensorFlow Lite); achieved quasi-real-time throughput (under 60 seconds per layer) enabling automated print pausing to reduce material waste by eliminating entire failed prints

Technical Skills

- **LLM & Generative AI:** Prompt Engineering, RAG, LangChain, LangGraph, Agentic AI, Function Calling, LLM Fine-tuning (LoRA/QLoRA), Hugging Face Transformers, Pinecone, pgvector, FAISS, Semantic Search
- **ML / Deep Learning:** PyTorch, TensorFlow, Scikit-learn, CNNs, Transformers, GANs (Pix2Pix), YOLOv5/YOLOv8, Transfer Learning, Reinforcement Learning (basics), Hyperparameter Optimisation

- ▶ **MLOps & Production AI:** Docker, Kubernetes (basics), MLflow, CI/CD for ML, Model Monitoring, Drift Detection, A/B Testing, FastAPI, REST APIs, Production Observability
- ▶ **Computer Vision:** OpenCV, Object Detection, Segmentation, Classification, Real-Time Video Processing, Orthorectification, Geospatial Image Analysis
- ▶ **Edge & Embedded AI:** TensorFlow Lite, Quantisation, Pruning, Embedded Inference, Raspberry Pi, Embedded Linux
- ▶ **Data Engineering & IoT:** Python, SQL, PostgreSQL, MongoDB, GPS/GNSS, LTE Modules, UART, Sensor Data Pipelines, Apache Kafka (basics), ThingsBoard
- ▶ **Programming & Tools:** Python, C/C++, Java, JavaScript, Git, GitHub Actions, Linux, Power BI, Jupyter, VS Code
- ▶ **Specialised Domains:** Geospatial AI (LiDAR, SAR, GPR), Multimodal Data Fusion, IoT Fleet Telemetry, Embedded Computer Vision

Education

Symbiosis Institute of Technology, Pune

2021 – 2025

B.Tech in Artificial Intelligence & Machine Learning

- ▷ Relevant Coursework: Deep Learning, Computer Vision, NLP, Reinforcement Learning, Data Structures & Algorithms, Cloud Computing, Big Data Analytics, Explainable AI, Cyber-Physical Systems, IoT, Embedded AI, Database Management Systems, Probability & Statistics for ML
- ▷ Highlights: Six-month industry internship; collaborated with Tech Mahindra Makers Lab on NLP & Computer Vision modules; NBA-accredited programme, NEP-2020 aligned curriculum; NIRF Rank 46 (Engineering, 2025)

Certifications

- ▶ IBM – Python for Data Science and AI (Coursera)
- ▶ IBM – Data Analytics and Visualization Capstone (Coursera)
- ▶ IBM – SQL for Data Science (Coursera)